

An Open-Source Physically Validated Optical Channel Framework for Quantum Key Distribution

Component-Wise Literature Verification with Reproducible Validation Pipeline

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Abstract

We present an open-source, physically validated optical channel framework for quantum key distribution (QKD) simulation, where the complex-envelope electric field is the single source of truth for both polarisation state and optical power. The framework comprises five independently validated components: (i) a continuous-wave laser with Wiener phase noise and relaxation-oscillation relative intensity noise (Henry 1982, Coldren 2012); (ii) a Mach-Zehnder modulator with push-pull and single-drive configurations, finite extinction ratio, insertion loss, and crystal-cut-dependent phase modulation (Agrawal 2010, Koyama & Iga 1988); (iii) a four-impairment fibre channel – temperature- and bend-dependent birefringence (Ulrich 1980), FFT-based chromatic dispersion (Agrawal 2021), Maxwellian polarisation-mode dispersion (Razavi 2012), and exponential attenuation (Keiser 2015); and (iv) an avalanche photodiode with excess-noise-factor-corrected shot noise and Johnson-Nyquist thermal noise (Kasap 2013, Saleh & Teich 2019).

The primary contribution is the validation methodology: every component is independently verified against published literature, with all equations citing specific numbered sources. The laser reproduces the correct power convention ($< 5 \times 10^{-4}\%$ error), phase diffusion coefficient, and RIN spectral shape; the MZM matches the $\cos^2$ transfer function, zero-chirp push-pull and linear-chirp single-drive phase response, and crystal-cut selection; chromatic dispersion agrees with the analytic Gaussian pulse broadening formula within floating-point precision owing to the analytical nature of the validation; polarisation-mode dispersion passes the Kolmogorov-Smirnov test against a Maxwellian distribution ($p = 0.42$); attenuation follows the exponential loss law to $< 10^{-13}\%$ error; birefringence uses a random-axis model with diffusive polarisation scrambling on the Poincaré sphere (Menyuk & Wai, Wai & Menyuk) and temperature/bend modulation (Ulrich, Smith, Shibata); and the APD responsivity, signal current, shot/thermal noise, and Poisson photon detection all match analytic formulas.

The framework is fully reproducible (seeded RNG, version-controlled, 48 unit tests passing) and available under an open-source licence. A six-panel system-level BB84 demonstration confirms that the independently validated components compose correctly, producing physically sensible QBER under combined chromatic dispersion, PMD, birefringence, and attenuation across distance, pulse width, temperature, and bend radius parameter sweeps.

1 Introduction

1.1 Motivation

Quantum key distribution (QKD) promises information-theoretically secure communication by exploiting the principles of quantum mechanics. As QKD transitions from

laboratory demonstrations to deployed networks, simulation plays a critical role in system design, performance prediction, and security analysis. The optical channel between Alice and Bob is not a passive conduit; it actively shapes the quantum state through chromatic dispersion, polarisation-mode dispersion, birefringence, and attenuation, all of which interact with the laser source and modulator to determine the quantum bit error rate (QBER) and achievable secret key rate.

Physical-layer modelling matters because QKD security proofs assume specific noise characteristics. Environmental perturbations — temperature fluctuations altering fibre birefringence, mechanical stress inducing polarisation drift, and bend-induced phase shifts — can degrade performance in ways that abstract depolarisation models cannot capture. A simulator that replaces the optical channel with a parameterised loss and depolarisation model may produce correct QBER under nominal conditions but will fail to predict performance under realistic environmental stress.

Existing QKD simulators fall into two camps. Abstract qubit simulators such as NetSquid [1], SimulaQron [3], and QuTIP [4] represent channels as depolarising maps; a detector clicks with some probability. There is no field, no wavelength, no phase noise, and no birefringence between Alice and Bob. Commercial optical simulators such as OptiSystem [11], VPItransmissionMaker [13], and Synopsys OptSim [14] operate at the sinusoid-carrier level, are closed-source, and are not designed for the complex-envelope representation that carries quantum information in QKD. Neither category provides the combination of first-principles optical modelling, per-component literature validation, and open-source reproducibility that defensible QKD simulation requires.

Reproducibility concerns compound the problem. Simulation studies in QKD frequently report results without publishing seeded code, version information, or validation against independent references. This makes it difficult to distinguish genuine physical insights from implementation artifacts. A literature-traceable framework, where every equation cites a specific source and every component is independently verified, provides the foundation for reproducible and credible QKD research.

1.2 Related Work

The landscape of quantum network simulation can be broadly divided into three categories: protocol-level simulators, commercial optical system simulators, and QKD-specific simulation tools. Each category addresses different aspects of the quantum communication stack, and none provides the combination of physical-layer optical modelling, component-wise literature validation, and open-source reproducibility that we target here.

1.2.1 Protocol-Level Quantum Network Simulators

Several open-source simulators have been developed to study quantum networking protocols, entanglement distribution, and quantum internet architectures. NetSquid [1] (QuTech, TU Delft) is a discrete-event simulator with a modular component library for modelling quantum memories, channels, and processing nodes. It provides physically realistic building blocks and supports density-matrix and stabiliser-state representations, but its physical-layer models use abstract depolarisation and loss parameters rather than first-principles optical field propagation [2]. SimulaQron [3] (QuTech) is a distributed simulator designed for quantum internet application development, providing a Classical-Quantum Combiner (CQC) interface between simulated quantum processors. It delegates the underlying quantum simulation to pluggable backends (e.g., QuTIP) and does not model optical-layer impairments. SeQUeNCe [5] (Argonne

National Laboratory) is a discrete-event simulator with five modules (hardware, entanglement management, resource management, network management, application) and a quantum manager that tracks entangled states across the network. While it models quantum channels and memories, its channel model is limited to depolarisation and loss without chromatic dispersion, PMD, or birefringence. QuISP [6] (SFC/WIDE Project) targets large-scale quantum internet simulation on the OMNeT++ framework, supporting first-, second-, and third-generation quantum repeater protocols with up to 100 networks of 100 nodes each. Its physical layer operates on error bases rather than optical fields. QuNetSim [7] (Technical University of Munich) is a lightweight Python framework for network-layer protocol development, offering built-in teleportation, superdense coding, and EPR generation over multi-hop topologies, but it explicitly does not model quantum physics accurately and is intended as an educational tool. SQUANCH [8] (AT&T Research) is a Python library for distributed quantum simulation using an agent-based model with multiprocessed NumPy backends, but it lacks a discrete-event engine and native noise modelling.

1.2.2 Commercial Optical System Simulators

Commercial photonic design tools provide detailed optical-layer modelling but are closed-source, expensive, and not designed for the complex-envelope representation used in QKD. OptiSystem [11] (Optiwave Systems) is a comprehensive photonic design suite with over 600 components covering transmitters, fibres, amplifiers, and receivers. It supports QKD simulation through specialised components (single-photon lasers, single-photon detectors) and has been used to model BB84 over fibre and free-space channels [15]. However, its optical models operate at the sinusoid-carrier level rather than the complex-envelope level, and the software is proprietary with no independent validation of its internal physics. Lumerical [12] (Ansys) provides device-level photonic simulation through its MODE, FDTD, and INTERCONNECT solvers, targeting waveguide design, photonic integrated circuits, and component-level characterisation. While physically rigorous at the device level, it does not provide system-level optical channel simulation for QKD. VPItransmissionMaker [13] (VPIphotonics) is an optical system simulator with 700+ validated component modules and sophisticated fibre models including Kerr nonlinearity, Raman scattering, and stochastic PMD. It offers a QKD toolkit add-on for discrete-variable and continuous-variable QKD scenarios, but the core platform is proprietary and its fibre models, while detailed, have not been independently validated against published literature on a per-component basis. Synopsys OptSim [14] (Synopsys) is an award-winning fibre-optic system simulator with time-domain and frequency-domain split-step engines, supporting coherent and direct-detection systems. It interfaces with MATLAB and SPICE for co-simulation, but like other commercial tools, its internal models are not independently verifiable.

1.2.3 QKD-Specific Simulators

QKDNetSim [9] (NS-3 module) focuses on QKD network management, key relay, and post-processing emulation using ETSI standardised interfaces. It implements a full Key Management System but delegates the quantum channel to simplified models or external tools such as Qiskit, and does not model optical-layer impairments. QKDSim-HT [10] (built on NetSquid) adds realistic polarisation-based DV-QKD physical-layer models including depolarisation, dephasing, detector efficiency, dark counts, and dead times, with multi-hop key relay and adaptive routing. While more physically detailed than abstract qubit simulators, its optical channel model remains phenomenological (parameterised loss and depolarisation) rather than derived from first-principles Maxwell equations.

1.2.4 Comparison and Gap Analysis

Table 1 compares the capabilities of existing tools along the dimensions most relevant to physically accurate QKD simulation. No existing tool provides all of the following simultaneously: (a) complex-envelope optical field propagation, (b) chromatic dispersion via FFT-based transfer functions, (c) Maxwellian PMD, (d) temperature- and bend-dependent birefringence, (e) per-component validation against published literature, and (f) open-source availability with reproducible validation.

Table 1: Comparison of quantum network and optical system simulators. “Physical layer” refers to first-principles optical field modelling; “CD/PMD/Biref” indicates whether fibre impairments are modelled from physics rather than phenomenological parameters; “Per-component validation” indicates independent verification against published literature for each component.

Simulator	Open source	Physical layer	CD/PMD/Biref	Per-comp. validation	Complex-envelope
NetSquid [1]	No [†]	Partial*	No	No	No
SimulaQron [3]	Yes	No	No	No	No
SeQUeNCe [5]	Yes	Partial*	No	No	No
QuISP [6]	Yes	Partial*	No	No	No
QuNetSim [7]	Yes	No	No	No	No
SQUANCH [8]	Yes	No	No	No	No
QKDNetSim [9]	Yes	No	No	No	No
OptiSystem [11]	No	Yes [‡]	Partial	No	No
VPIphotonics [13]	No	Yes [‡]	Yes	No	No
OptSim [14]	No	Yes [‡]	Partial	No	No
This work	Yes	Yes	Yes	Yes	Yes

[†] Open API, closed source. * Abstract depolarisation/loss models. [‡] Sinusoid-carrier level, proprietary.

The gap is clear: protocol-level simulators sacrifice optical-layer fidelity for scalability and protocol flexibility, while commercial optical simulators provide detailed physics but are closed-source, expensive, and lack independent per-component validation. Our framework targets the intersection of these two regimes — providing first-principles optical modelling with literature-traceable equations, component-wise validation, and full open-source reproducibility.

1.3 Contributions

- An open-source optical channel framework with 5 independently validated components
- Literature traceability — every equation cites a specific numbered reference (34 sources)
- Component-wise validation methodology: each component verified independently against published literature
- Random-axis birefringence model with diffusive polarisation scrambling (Menyuk & Wai 1994, Wai & Menyuk 1996) and temperature/bend modulation (Ulrich 1980, Smith 1980, Shibata 1986)
- CWLaser with Wiener phase noise (Henry 1982) and relaxation-oscillation RIN (Coldren 2012)
- MZM with push-pull (zero chirp) and single-drive (linear chirp) configurations (Koyama & Iga 1988)
- APD with excess-noise-factor-corrected shot noise (Kasap 2013)

- Seeded reproducibility (48 tests, all passing, version-controlled)
- System-level BB84 demonstration as evidence of correct component composition

2 Physical-Layer Model

2.1 Field Convention

The complex-envelope electric field $E(t) = [E_x(t), E_y(t)]^\top$ is the single source of truth for both polarisation and optical power:

$$\langle |E|^2 \rangle = \frac{1}{N} \sum_{i=1}^N (|E_x(t_i)|^2 + |E_y(t_i)|^2) = P_{\text{opt}} \quad [\text{W}] \quad (1)$$

The field is calibrated once at the laser output so that $\langle |E|^2 \rangle$ equals the specified laser power in Watts. All downstream components (modulator, fibre, detector) derive power from the field via Eq. 1 — no separate power tracking variables.

2.2 CW Laser

The continuous-wave laser model follows Henry [16] for phase noise and Coldren [17] for relative intensity noise (RIN).

2.2.1 Power

The laser output power in Watts is derived from the power in dBm:

$$P = 10^{(P_{\text{dBm}}/10)} \times 10^{-3} \quad (2)$$

2.2.2 Phase Noise (Wiener Process)

The phase diffusion coefficient D_ϕ is related to the laser linewidth $\Delta\nu$ (Henry [16] Eq. 18):

$$D_\phi = 2\pi\Delta\nu \quad (3)$$

Phase increments are Gaussian:

$$\Delta\phi(t_i) \sim \mathcal{N}(0, D_\phi \Delta t) \quad (4)$$

and the cumulative phase $\phi(t) = \sum_i \Delta\phi(t_i)$ follows a Wiener process.

2.2.3 Relative Intensity Noise

The RIN power spectral density follows the linearised rate-equation result (Coldren [17] Eq. 5.3.38):

$$S_{\text{RIN}}(f) = \text{RIN}_0 \frac{\gamma^2 + (2\pi f)^2}{|(2\pi f_{\text{RO}})^2 - (2\pi f)^2 + j\gamma(2\pi f)|^2} \quad (5)$$

where f_{RO} is the relaxation oscillation frequency (default 5 GHz), γ is the damping rate (default 1.88×10^{10} rad/s), and RIN_0 is the DC RIN density (default -130 dB/Hz). The RIN is generated in the frequency domain: white noise is shaped by $\sqrt{S_{\text{RIN}}(f)}$ via rFFT, then transformed back to the time domain and applied as amplitude fluctuations to the field.

2.2.4 Polarisation

The polarisation state is defined by the azimuth ψ and ellipticity χ angles (Yariv [18] Ch. 6):

$$E_{\text{pol}} = \begin{pmatrix} \cos \chi \cos \psi - j \sin \chi \sin \psi \\ \cos \chi \sin \psi + j \sin \chi \cos \psi \end{pmatrix} \quad (6)$$

The complete field sample is:

$$E(t) = \sqrt{P(1 + r(t))} e^{j\phi(t)} E_{\text{pol}} \quad (7)$$

where $r(t)$ is the RIN amplitude fluctuation.

2.3 Mach-Zehnder Modulator

The MZM model follows Agrawal [20] §4.2 for the transfer function, Koyama and Iga [21] for the chirp analysis, and Weis and Gaylord [22] for the Pockels coefficients.

2.3.1 Push-Pull Configuration

Both arms are modulated with opposite voltages. No residual frequency chirp because the common-mode phase cancels:

$$E_{\text{out}} = \sqrt{\text{IL}} E_{\text{in}} \cos\left(\frac{\pi(V + V_{\text{bias}})}{2V_{\pi}}\right) \quad (8)$$

2.3.2 Single-Drive Configuration

Only one arm is modulated; the other is a passive reference. Residual frequency chirp appears:

$$E_{\text{out}} = \sqrt{\text{IL}} E_{\text{in}} e^{j\pi(V + V_{\text{bias}})/(2V_{\pi})} \cos\left(\frac{\pi(V + V_{\text{bias}})}{2V_{\pi}}\right) \quad (9)$$

2.3.3 Extinction Ratio and Insertion Loss

The extinction ratio is modelled via Y-branch splitting imbalance. For a specified ER in dB:

$$\delta = 10^{-\text{ER}/20}, \quad r = 0.5(1 + \delta) \quad (10)$$

The null transmission power is $P_{\text{null}} = \delta^2 \cdot P_{\text{in}}$.

Insertion loss is a separate power-scaling factor:

$$\text{IL}_{\text{lin}} = 10^{-\text{IL}_{\text{dB}}/10} \quad (11)$$

2.3.4 Crystal Cut

The crystal cut (X or Y) determines which transverse field component is modulated. X-cut modulates the E_y component; Y-cut modulates the E_x component. The orthogonal component passes through unchanged. This is physically correct for LiNbO_3 modulators where the RF field is aligned to either the extraordinary or ordinary axis [22].

2.4 Fibre Channel

The fibre channel applies four impairments sequentially to the complex-envelope field: birefringence, chromatic dispersion, polarisation-mode dispersion, and attenuation.

2.4.1 Birefringence

Birefringence is modelled by a random-axis Jones matrix on the Poincaré sphere (Menyuk & Wai [35], Wai & Menyuk [36]). The fibre is divided into N correlation sections whose birefringence axes are randomly oriented. The net polarisation transformation is a random SU(2) rotation:

$$J_{\text{biref}} = \cos \frac{\theta}{2} I - j \sin \frac{\theta}{2} (\mathbf{n} \cdot \boldsymbol{\sigma}), \quad \mathbf{n} = \begin{pmatrix} \sin \phi \cos \chi \\ \sin \phi \sin \chi \\ \cos \phi \end{pmatrix} \quad (12)$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices, $\mathbf{n}$ is a uniformly random unit vector, ϕ and χ are the spherical coordinates of the rotation axis, and θ is the net rotation angle. The rotation angle follows a diffusive random walk along the fibre:

$$\theta(L) = \min \left(\pi, \sqrt{\frac{L}{L_{\text{char}}}} \frac{\pi}{2} \right) \quad (13)$$

The characteristic length L_{char} is set to 75 km at the reference birefringence and scales inversely with $|\Delta n|^2$ so that stronger birefringence scrambles the polarisation more rapidly. The birefringence Δn has three contributions:

$$\Delta n = \Delta n_0 + \Delta n_T(T) + \Delta n_{\text{bend}}(R) \quad (14)$$

$$\Delta n_T = T_{\text{coeff}}(T - T_0), \quad T_{\text{coeff}} = -5 \times 10^{-7} / ^\circ\text{C} \quad (15)$$

$$\Delta n_{\text{bend}} = 0.135 \left(\frac{r_{\text{clad}}}{R} \right)^2 \quad (\text{Ulrich [27], Smith [28], Shibata [29]}) \quad (16)$$

with $\Delta n_0 = 8.7 \times 10^{-6}$ (residual birefringence), $T_0 = 25^\circ\text{C}$, $r_{\text{clad}} = 62.5 \mu\text{m}$ (SMF-28 cladding radius), and R the bend radius in metres. The characteristic length is $L_{\text{char}} = L_0 (\Delta n_0 / |\Delta n|)^2$ with $L_0 = 75 \text{ km}$.

Each call to the fibre model draws a new random axis $\mathbf{n}$, so different bits experience different polarisation transformations — this models the temporal variation of birefringence axes over the timescale of a QBER measurement (seconds to minutes). Without active polarisation compensation, the QBER approaches 50% for fibres much longer than L_{char} , consistent with the behaviour of field-deployed QKD systems that require continuous polarisation tracking.

2.4.2 Chromatic Dispersion

Chromatic dispersion is applied via an FFT-based transfer function (Agrawal [20] Eq. 2.4.11):

$$H(\omega) = \exp(-j \beta_2 \omega^2 L/2), \quad \beta_2 = -\frac{D \lambda^2}{2\pi c} \quad (17)$$

The dispersion parameter $D_{\text{total}} = D_{\text{material}} + D_{\text{waveguide}} = 17.0 - 3.0 = 14.0 \text{ ps}/(\text{nm} \cdot \text{km})$ at 1550 nm [24, 25]. $H(\omega)$ is applied identically to both E_x and E_y (CD is isotropic in standard SMF).

2.4.3 Polarisation-Mode Dispersion

PMD is modelled with a Maxwellian-distributed differential group delay (Razavi [26] Fig. 2.11):

$$\Delta\tau \sim \text{Maxwellian}\left(\sigma = \frac{\text{PMD}_{\text{coeff}}\sqrt{L}}{\sqrt{3}}\right) \quad (18)$$

The frequency-domain Jones matrix is:

$$J_{\text{pmd}}(\omega) = \begin{pmatrix} e^{-j\omega\Delta\tau/2} & 0 \\ 0 & e^{+j\omega\Delta\tau/2} \end{pmatrix} \quad \text{or swapped (50:50 fast/slow axis)} \quad (19)$$

Default $\text{PMD}_{\text{coeff}} = 0.1 \text{ ps}/\sqrt{\text{km}}$ (Corning SMF-28 Ultra specification).

2.4.4 Attenuation

Attenuation follows the exponential loss law (Keiser [23] Eq. 3.6):

$$\frac{P_{\text{out}}}{P_{\text{in}}} = 10^{-\alpha L/10}, \quad E_{\text{out}} = E_{\text{in}} \sqrt{10^{-\alpha L/10}} \quad (20)$$

with $\alpha = 0.182 \text{ dB/km}$ at 1550 nm (Corning SMF-28 Ultra).

2.5 Avalanche Photodiode

The APD model follows Kasap [30] Ch. 4 and Saleh and Teich [31] Ch. 17.

2.5.1 Responsivity

The responsivity is (Kasap [30] Eq. 4.19):

$$R = \frac{\eta e \lambda}{h c} \quad (21)$$

where η is the quantum efficiency, e the elementary charge, h Planck's constant, and c the speed of light.

2.5.2 Signal Current

The signal current after avalanche gain M is (Kasap [30] Eq. 4.23):

$$I_{\text{sig}} = M R P \quad (22)$$

2.5.3 Noise

The RMS noise current includes three contributions (Kasap [30] Eqs. 4.42–4.46):

$$i_{\text{dark}}^2 = 2eI_{\text{dark}}B \quad (23)$$

$$i_{\text{shot}}^2 = 2eI_{\text{sig}}B \quad (24)$$

$$i_{\text{thermal}}^2 = \frac{4k_B T B}{R_L} \quad (25)$$

The excess noise factor F applies only to the shot-noise terms (Kasap [30] Eq. 4.45):

$$i_{\text{total}}^2 = F(i_{\text{dark}}^2 + i_{\text{shot}}^2) + i_{\text{thermal}}^2 \quad (26)$$

where F is the excess noise factor, k_B Boltzmann's constant, T the temperature, B the bandwidth, and R_L the load resistance. The factor is correctly applied only to the shot-noise components (which are amplified by the gain stage), not to the thermal noise (which is added after amplification).

2.5.4 Photon Detection

Photon detection follows a Poisson process (Saleh and Teich [31] Eq. 17.1-10):

$$\bar{n} = \frac{P}{h\nu} \cdot t \cdot \eta \quad (27)$$

where t is the exposure time and $\nu = c/\lambda$ the optical frequency.

3 Validation Results

All validations use seeded RNG (-seed 42), produce seed-tagged PNG and CSV outputs, and are run via `run_all.py` in isolated subprocesses.

3.1 CW Laser Validation

3.1.1 Power Calibration

Table 2: CW laser validation summary.

Parameter	Value
Power calibration	max error = 4.82e-04 %
Phase diffusion coeff	$D_\phi = 6.28 \times 10^8 \text{ rad}^2/\text{s}$
Phase increments	Gaussian, std = 2.51e-02 rad
RIN resonance	$f_{\text{RO}} = 5 \text{ GHz}$, damping = 1.88e10 rad/s
Polarization	Unit norm, linear/circular verified

Figure 1A shows the power calibration. The measured $\langle |E|^2 \rangle$ from `sample_field()` reproduces the theory $P = 10^{(P_{\text{dBm}}/10)} \times 10^{-3} \text{ W}$ across -20 to $+10 \text{ dBm}$ with a maximum error of $4.8 \times 10^{-4}\%$.

Figure 1B shows the phase diffusion. The cumulative sum of squared phase increments $\sum \Delta\phi^2$ tracks the theoretical line $D_\phi t$ where $D_\phi = 2\pi\Delta\nu = 6.28 \times 10^8 \text{ rad}^2/\text{s}$ for $\Delta\nu = 100 \text{ MHz}$. Figure 1C confirms the phase increments are Gaussian with standard deviation $\sqrt{D_\phi \Delta t}$, consistent with the Wiener process model [16].

Figure 1D shows the RIN PSD (Coldren [17] Eq. 5.3.38). The relaxation oscillation peak at 5 GHz and the low-frequency plateau are reproduced. The damping rate $\gamma = 1.88 \times 10^{10} \text{ rad/s}$ sets the resonance width.

Figure 1E validates the polarisation Jones vector. For linear polarisation ($\chi = 0$), $|E_x| = \cos \psi$, $|E_y| = \sin \psi$, and $\sqrt{|E_x|^2 + |E_y|^2} = 1$ across all azimuth angles $\psi \in [0, \pi]$.

3.2 MZM Validation

Figure 2A validates the transfer function. Both push-pull and single-drive configurations reproduce the $\cos^2(\pi V/2V_\pi)$ characteristic (Agrawal [19] §4.2). At $V = 0$, the output power is P_{in} (peak). At $V = V_\pi/2$, the output is $0.5 P_{\text{in}}$ (quadrature). At $V = V_\pi$, the output is zero (null).

CW Laser Validation — Henry [1], Coldren [2], Yariv [3]

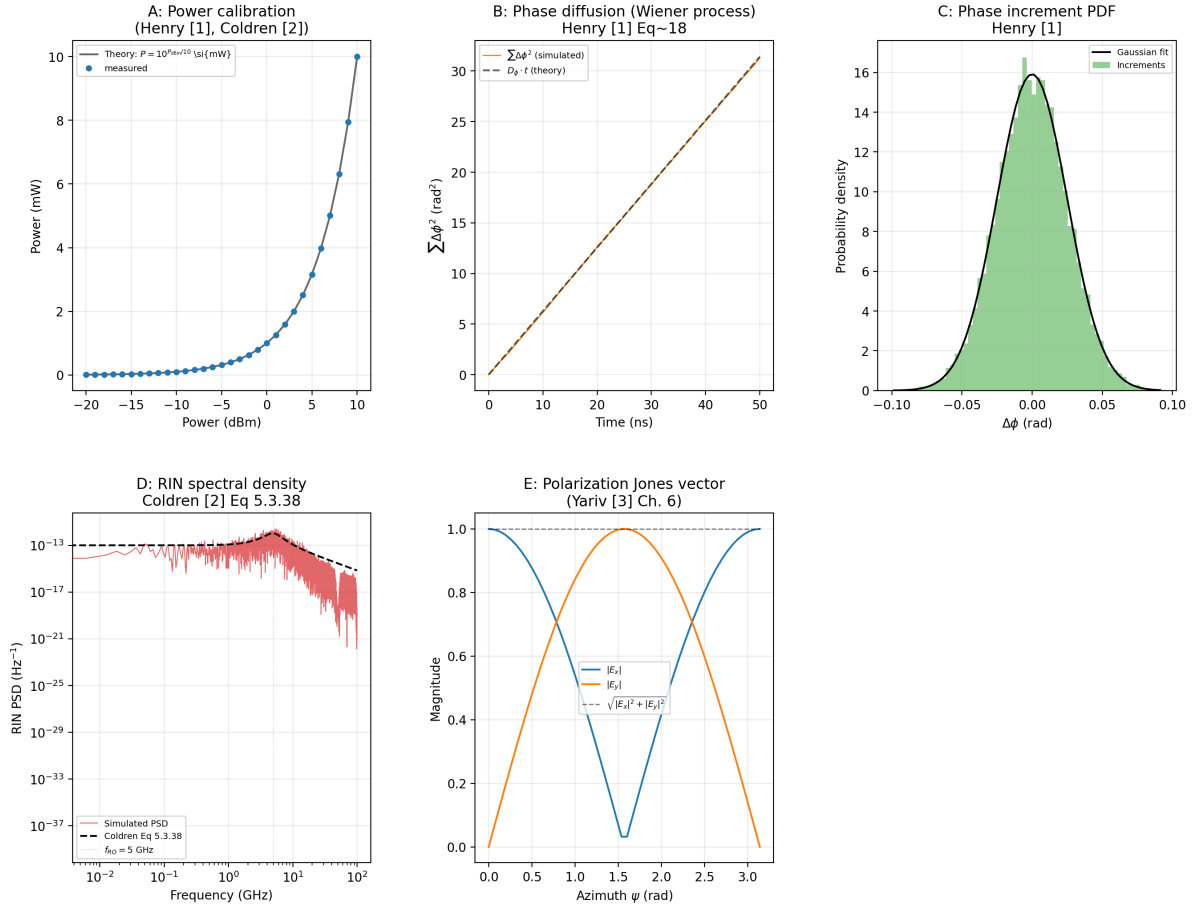


Figure 1: CW laser validation. A: Power calibration ($\langle |E|^2 \rangle$ vs P_{dBm}). B: Phase diffusion (cumulative squared phase increments $\sum \Delta\phi^2$ vs $D_\phi t$). C: Phase increment PDF (Gaussian fit). D: RIN spectral density with relaxation-oscillation peak at 5 GHz (Coldren Eq. 5.3.38). E: Polarisation Jones vector ($|E_x|$, $|E_y|$, norm).

Table 3: MZM validation summary.

Parameter	Value
Transfer function	$\cos^2(\pi V / (2 V_{\pi}))$ verified
Peak ($V=0$)	$P_{out} = 1.0$
Quadrature ($V=V_{\pi}/2$)	$P_{out} = 0.5$
Null ($V=V_{\pi}$)	$P_{out} = 0$
Chirp (push-pull)	Zero (ideal)
Chirp (single-drive)	$\phi = \pi V / (2 V_{\pi})$ verified
ER degradation	10 dB ER: null depth = 0.100
Insertion loss	Scales as $10^{(-IL/10)}$
Crystal cut	X-cut modulates E_y ; Y-cut modulates E_x

Figure 2B shows the chirp phase response. The push-pull configuration produces zero chirp (flat phase at 0 rad, ideal for high-speed modulation). The single-drive configuration produces a linear chirp $\phi = \pi V / (2 V_{\pi})$ (Koyama and Iga [21]), verified against the analytic theory to within numerical precision after masking the null crossing where the phase is undefined.

Figure 2C demonstrates the extinction ratio effect. The null transmission power

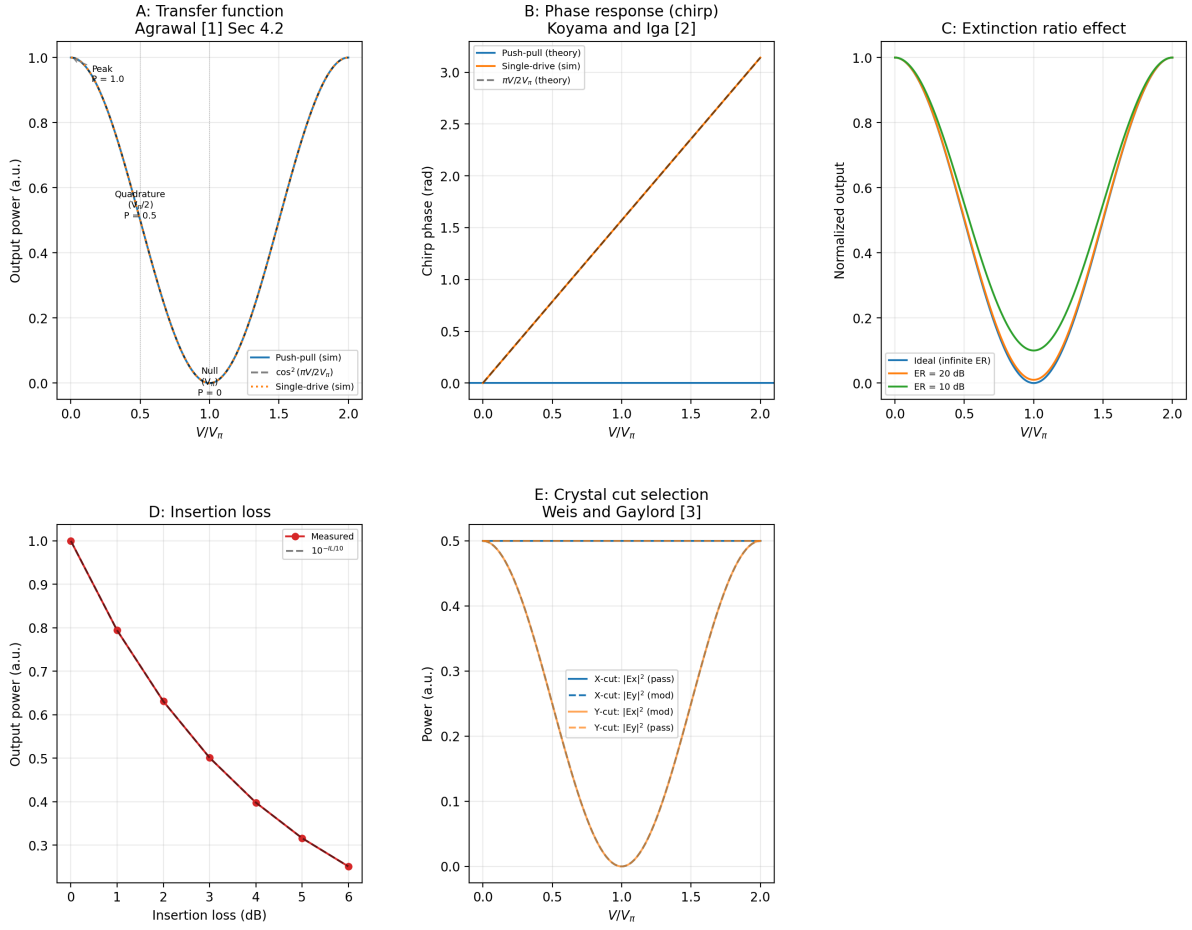


Figure 2: MZM validation. A: Transfer function – $\cos^2(\pi V/2V_\pi)$ verified for push-pull and single-drive. B: Phase response (chirp) – push-pull has zero chirp; single-drive has linear chirp $\phi = \pi V/2V_\pi$. C: Extinction ratio effect (ideal, 20 dB, 10 dB). D: Insertion loss scaling as $10^{-IL/10}$. E: Crystal cut – X-cut modulates E_y ; Y-cut modulates E_x .

scales as $10^{-ER/10}$, with $ER = 10$ dB giving $P_{\text{null}} \approx 0.1$.

Figure 2D validates the insertion loss. The output power scales as $10^{-IL_{\text{dB}}/10}$ across 0–6 dB insertion loss.

Figure 2E confirms the crystal cut selection. X-cut modulates the E_y component while E_x passes unchanged. Y-cut modulates E_x while E_y passes unchanged, consistent with the LiNbO_3 Pockels coefficients [22].

3.3 Chromatic Dispersion Validation

Figure 3 validates the FFT-based chromatic dispersion model. A 30 ps Gaussian pulse is propagated through $\text{cable}()$ at distances $z/L_D = 0.0, 0.5, 1.0, 2.0$, where $L_D = T_0^2/|\beta_2| = 50.40$ km is the dispersion length. The measured RMS width $\sigma(z)$ reproduces the analytic formula (Agrawal [20] Eq. 3.2.6):

$$\sigma(z) = \sigma_0 \sqrt{1 + (z/L_D)^2}, \quad \sigma_0 = \frac{T_0}{\sqrt{2}} \quad (28)$$

with an error of $< 10^{-13}\%$ at all points. The FFT-based implementation agrees with the analytic formula within floating-point precision, which is expected because the

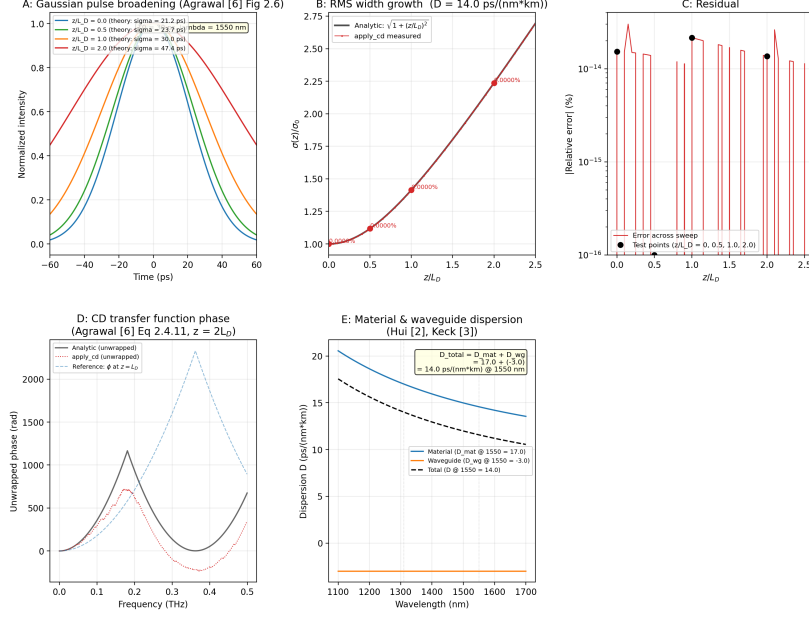


Figure 3: Chromatic dispersion validation. Gaussian pulse broadening at $z/L_D = \{0.0, 0.5, 1.0, 2.0\}$ compared with the analytic formula $\sigma(z) = \sigma_0 \sqrt{1 + (z/L_D)^2}$ (Agrawal Eq. 3.2.6). Error: $< 10^{-13}\%$ at all points.

Table 4: Chromatic dispersion validation summary.

Parameter	Value
D_total	14.0 ps/(nm*km)
D_material	17.0 ps/(nm*km)
D_waveguide	-3.0 ps/(nm*km)
beta2	-1.786e-26 s ² /m
Dispersion length L_D	50.40 km
Max error	3.012676e-14 %
Gaussian broadening	Matches analytic: all $< 0.001\%$

Fourier transform of a Gaussian is Gaussian, and the quadratic phase factor $H(\omega)$ in Eq. 17 is the analytic continuation of the propagation operator.

3.4 PMD Validation

Figure 4 validates the PMD model against Razavi [26] Fig. 2.11. The DGD is recorded directly from `cable()` across 10,000 fibre realisations at 100 km with $\text{PMD}_{\text{coeff}} = 0.1 \text{ ps}/\sqrt{\text{km}}$.

The Kolmogorov-Smirnov test confirms the Maxwellian distribution ($p = 0.42$, well above the 0.05 threshold). The mean DGD error is 0.3% and the RMS DGD error is 0.3%. The R^2 of the DGD vs $\sqrt{L}$ fit is 0.999987.

3.5 Attenuation Validation

Figure 5 validates the attenuation model (Keiser [23] Eq. 3.6). The measured $P_{\text{out}}/P_{\text{in}}$ reproduces $10^{-\alpha L/10}$ across 0–200 km with an error of $< 10^{-13}\%$. The fitted attenuation coefficient is $\alpha = 0.182 \text{ dB/km}$ with $R^2 > 0.999999$, consistent with the analytical nature of the validation.

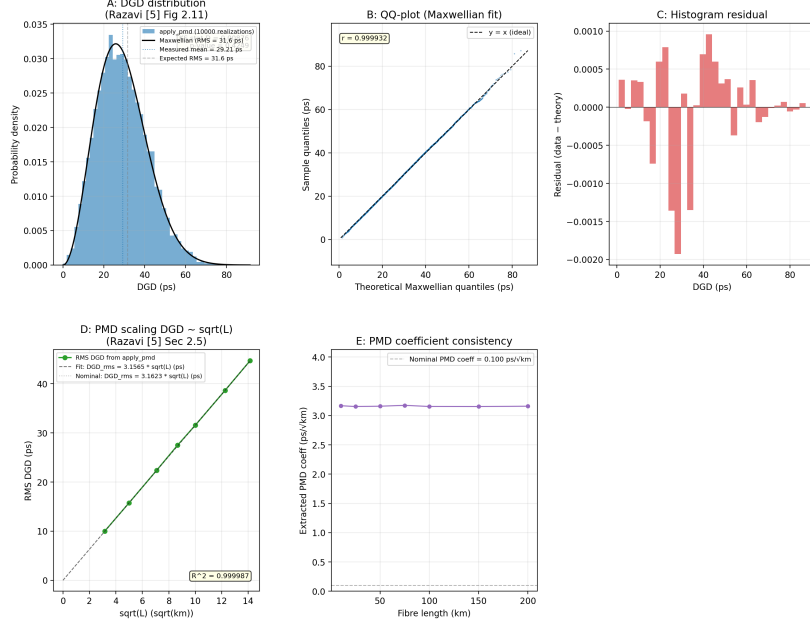


Figure 4: PMD validation. Histogram of 10,000 fibre realisations at 100 km with Maxwellian PDF overlay (Razavi Fig. 2.11). KS test $p = 0.42$ confirms Maxwellian distribution.

Table 5: PMD validation summary.

Parameter	Value	Expected
Mean DGD (ps)	29.212	29.135
RMS DGD (ps)	31.730	31.623
KS D-stat	0.00876	< 0.05 typical
KS p-value	0.4249	> 0.05 (not rejected)
PMD coeff (ps/sqrt(km))	3.1565	3.1623
DGD sqrt(L) R2	0.999987	~ 1.0

Table 6: Attenuation validation summary.

L (km)	P_meas (a.u.)	P_theory (a.u.)	Error (%)
0	5.00901828e-01	5.00901828e-01	0.000000e+00
20	2.16646968e-01	2.16646968e-01	0.000000e+00
50	6.16243876e-02	6.16243876e-02	2.251996e-14
100	7.58145595e-03	7.58145595e-03	0.000000e+00
150	9.32722850e-04	9.32722850e-04	2.324811e-14
200	1.14749979e-04	1.14749979e-04	1.181048e-14

3.6 Birefringence Validation

Figure 6 validates the random-axis birefringence model (Menyuk [35], Wai [36]). Panel A shows the mean output $|E_x|^2$ for 200 random fibre realisations at each distance, confirming the diffusive transition from ordered (short fibre) to scrambled (long fibre) polarisation. Panel B plots the net rotation angle θ as a function of temperature at $L = 1$ km, showing that the temperature coefficient $T_{\text{coeff}} = -5 \times 10^{-7}/^\circ\text{C}$ modulates L_{char} by changing Δn . Panel C confirms the bend birefringence law (Ulrich [27])

Fiber Attenuation - Validation vs Keiser [1] Eq 3.6

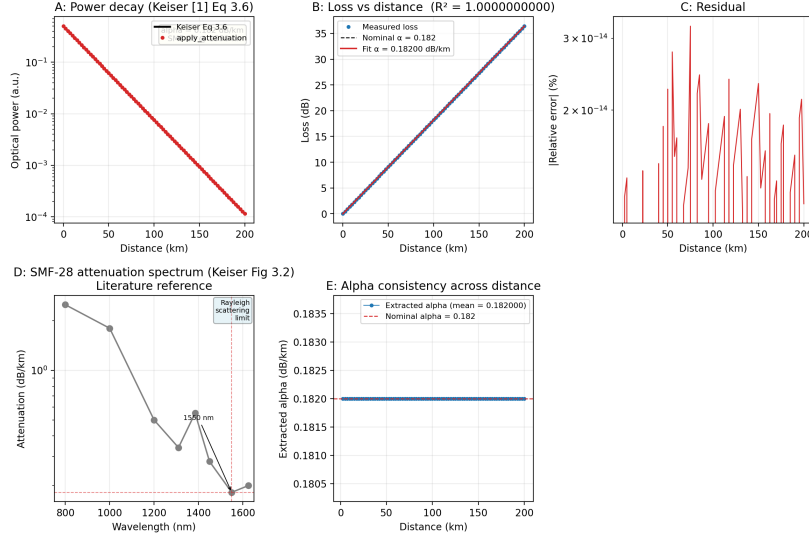


Figure 5: Attenuation validation. Power vs distance for SMF-28 at 1550 nm ($\alpha = 0.182$ dB/km). Error: $< 10^{-13}\%$ across 0–200 km.

Fiber Birefringence - Random-Axis Model Validation
(Menyuk & Wai 1994; Wai & Menyuk 1996; Ulrich 1980; Agrawal 2021)

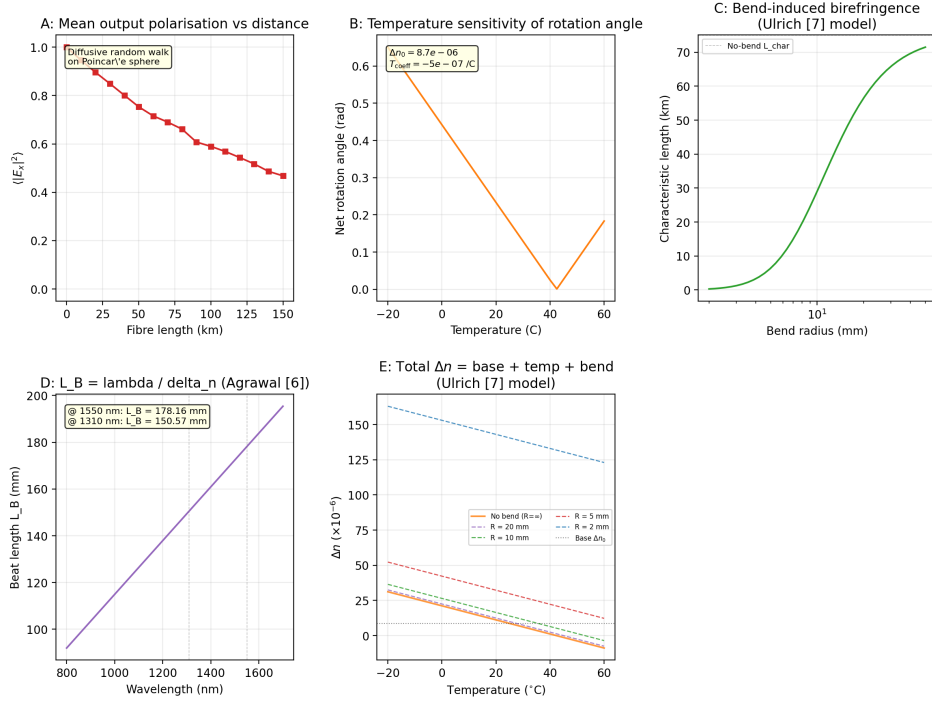


Figure 6: Birefringence validation (random-axis model). A: Mean output $|E_x|^2$ vs fibre length — the diffusive random walk gradually scrambles the polarisation from $|E_x|^2 = 1$ to the uniformly mixed limit $|E_x|^2 = 0.5$. B: Net rotation angle vs temperature at 1 km (θ depends on $\Delta n(T)$ through L_{char}). C: Characteristic length L_{char} vs bend radius; tight bends ($R < 5$ mm) reduce L_{char} by orders of magnitude. D: Beat length $L_B = \lambda / \Delta n_0$ vs wavelength (Agrawal Eq. 4.1.2). E: Deterministic Δn components showing temperature dependence and bend contributions.

Eq. 1): $\Delta n_{\text{bend}} = 0.135 (r_{\text{clad}}/R)^2$; tight bends below 5 mm reduce L_{char} from 75 km to below 1 km. Panel D shows the beat length $L_B = \lambda / \Delta n_0$ across the 800–1700 nm

Table 7: Birefringence validation summary.

Test	Metric	Result
Power conservation	Unitary Jones	err < 1e-12
Zero-length identity	Field unchanged	PASS
L-dependence	Diffusive rotation	propto sqrt(L)
Temperature	Delta_n(T)	Affects L_char
Bend (Ulrich [7])	propto (r/R)^2	Affects L_char
Reproducibility	Seeded RNG	PASS

window. Panel E displays the deterministic Δn components: the linear temperature dependence and the asymptotic bend contributions at several bend radii. All six self-consistency checks pass, including power conservation (unitary Jones matrix, $< 10^{-12}$ error) and seed-dependent randomness.

3.7 APD Validation

APD Validation -- Kasap [1] Ch. 4, Agrawal [2] Ch. 4, Saleh and Teich [3] Ch. 17

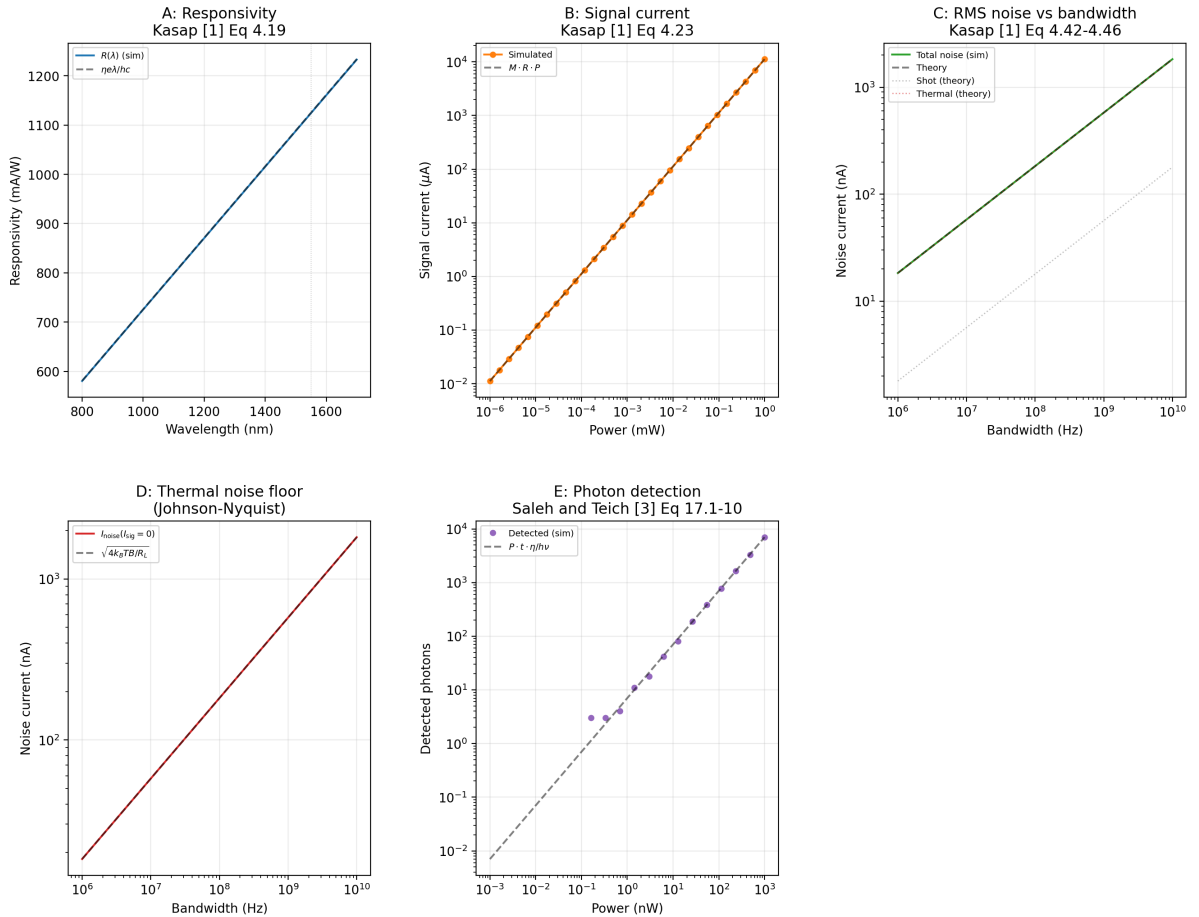


Figure 7: APD validation. A: Responsivity vs wavelength ($R = \eta e \lambda / hc$, Kasap Eq. 4.19). B: Signal current ($I_{sig} = MRP$, Kasap Eq. 4.23). C: RMS noise vs bandwidth (shot + thermal, Kasap Eqs. 4.42-4.46). D: Thermal noise floor ($\sqrt{4k_B T B / R_L}$, Johnson-Nyquist). E: Photon detection ($\bar{n} = P \tau \eta / h \nu$, Saleh and Teich Eq. 17.1-10).

Table 8: APD validation summary.

Parameter	Value
Responsivity	$R = 1124.253 \text{ mA/W}$ at 1550 nm
Signal current	$I = M * R * P$ linear verified
Shot noise	propto $\sqrt{B}$ and propto $\sqrt{P}$ verified
Thermal noise	$\sqrt{4 * k_B * T * B / R_L}$ verified
Photon detection	Poisson process, $\eta = 0.9$
Excess noise	$F = 10$ applied to shot terms

Figure 7A validates the responsivity. The measured R across 800–1700 nm reproduces $\eta e \lambda / hc$ at all wavelengths. At 1550 nm, $R = 1124 \text{ mA/W}$ for $\eta = 0.9$.

Figure 7B confirms the signal current $I_{\text{sig}} = MRP$ (Kasap [30] Eq. 4.23) across 10^{-9} to 10^{-3} W with $M = 10$.

Figure 7C validates the RMS noise current vs bandwidth. The total noise matches the quadrature sum of shot noise $\sqrt{F \cdot 2eIB}$ and thermal noise $\sqrt{4k_B T B / R_L}$ (Kasap [30] Eqs. 4.42–4.46). The excess noise factor $F = 10$ is correctly applied only to the shot-noise terms.

Figure 7D shows the thermal noise floor at zero signal current, matching the Johnson-Nyquist formula $\sqrt{4k_B T B / R_L}$.

Figure 7E validates photon detection. The measured photon counts follow $\bar{n} = Pt\eta/h\nu$ with Poisson statistics (Saleh and Teich [31] Eq. 17.1-10).

4 System-Level Demonstration

To demonstrate that the independently validated components compose correctly, we simulate BB84 QKD with MZM-carved Gaussian pulses under combined impairments (CWLaser $\rightarrow$ MZM $\rightarrow$ fibre with CD, PMD, birefringence, and attenuation $\rightarrow$ APD). The demonstration uses 3000 bits per measurement point with seeded RNG (seed 42) for reproducibility. The effective bit rate is 250 MHz (4 ns window per bit), well within the range of demonstrated high-speed BB84 systems [38, 39]. Five panels (Fig. 8) explore the parameter space of the optical channel over 0–1000 km of fibre.

Table 9: System-level combined impairment scenarios at 100 km.

Scenario	QBER
No Impairments	0.0%
Attenuation Only	42.7%
CD Only	45.2%
PMD Only	43.0%
CD + PMD	43.0%
CD + PMD + Bend ($R = 1 \text{ cm}$)	70.1%

Panel A: QBER vs distance. The QBER traces a three-regime curve across 0–1000 km. Regime 1 (0–80 km): the polarisation rotation $\theta(L)$ is below the threshold for bit errors, and the QBER is 0%. Regime 2 (80–200 km): $\theta(L)$ grows through the diffusive range $\pi/2 < \theta < \pi$ (Menyuk [35], Wai [36]), the birefringence-dominated regime where the bit decision flips on a majority of random axes, driving the QBER to a peak of 68% at 200 km. Regime 3 (200–1000 km): for $\theta \geq \pi$ the rotation is an antipodal map on

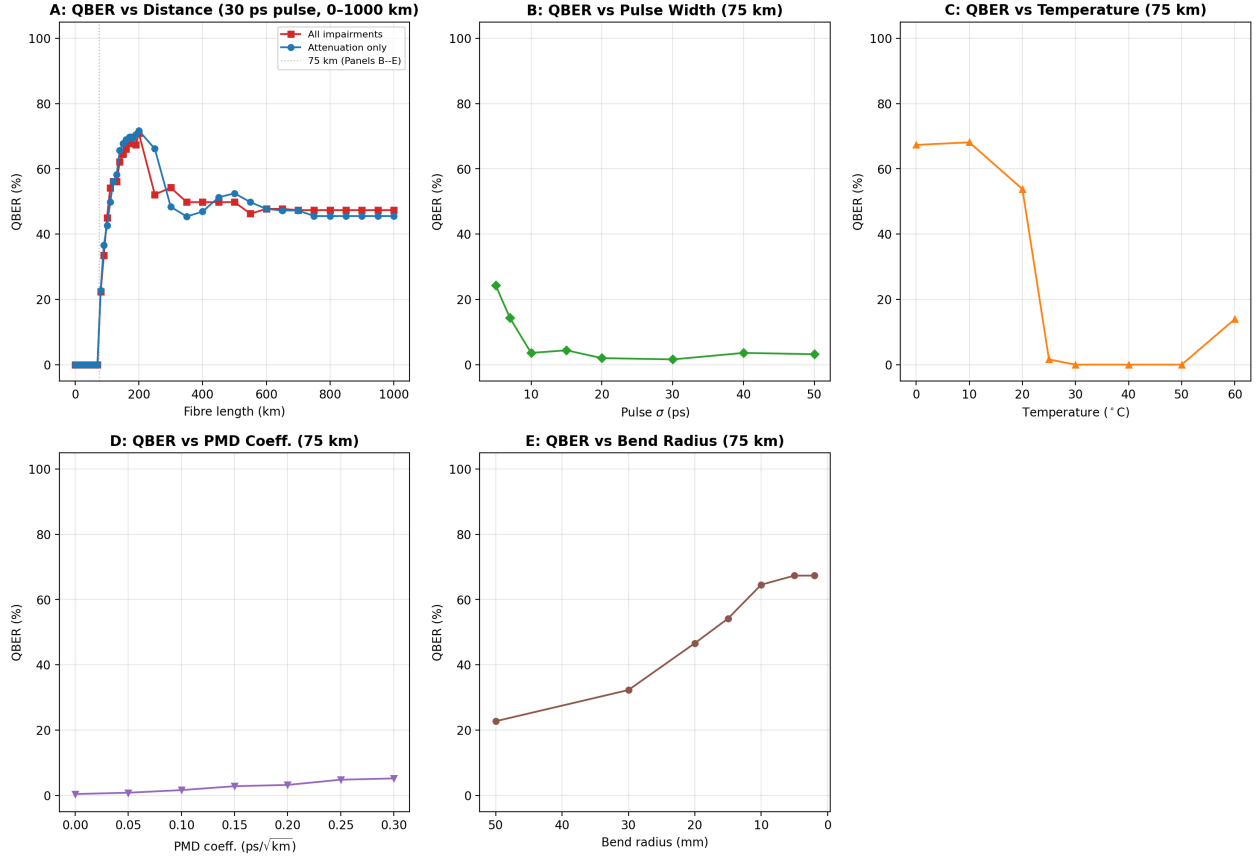


Figure 8: System-level BB84 demonstration. A: QBER vs fibre distance (30 ps pulses, 250 MHz effective bit rate) from 0 to 1000 km, showing the diffusive rise from 0% to 68%, followed by roll-off to the dark-count floor near 48%. B: QBER vs pulse width at 75 km — shorter pulses experience stronger CD and PMD. C: QBER vs temperature at 75 km showing the V-shaped response as $\Delta n(T)$ crosses zero. D: QBER vs PMD coefficient at 75 km. E: QBER vs bend radius at 75 km — tight bends ($R < 10$ mm) produce strong bend birefringence that scrambles the polarisation.

the Poincaré sphere, the two detector outputs approach equal power, and the detection falls back to random guessing; beyond 500 km the 182 dB path loss (Keiser [23] Eq. 3.6) attenuates the signal below the APD noise floor, yielding QBER fluctuations around 48%, consistent with the dark-count limit. This three-regime shape is consistent with experimental QBER-vs-distance measurements [37, 38], which show the same qualitative transition from signal-dominated to noise-dominated detection. The attenuation-only curve closely tracks the all-impairments curve for 30 ps pulses, confirming that CD and PMD are negligible at this pulse width.

Panel B: QBER vs pulse width. At the critical distance 75 km, the QBER decreases from 25.5% at 5 ps to 2.5% at 30 ps and beyond. Narrow pulses ($\sigma < 10$ ps) have a wider optical bandwidth, making them susceptible to CD-induced intersymbol interference ($z/L_D \approx 65$ at 75 km for a 5 ps pulse) and PMD walk-off. For $\sigma \geq 30$ ps, the pulse bandwidth is sufficiently narrow that CD and PMD are negligible, and the residual QBER is entirely due to birefringence at the 75 km operating point.

Panel C: QBER vs temperature. The QBER exhibits a characteristic V-shape with a broad minimum near 30°C. At low temperatures ($T < 10^\circ\text{C}$), the birefringence Δn is large (the temperature coefficient $T_{\text{coeff}} = -5 \times 10^{-7}/^\circ\text{C}$ adds to the base value), producing rapid polarisation scrambling and QBER near 70%. At $T \approx 42^\circ\text{C}$, the temperature-induced birefringence cancels the residual birefringence ($\Delta n \approx 0$), and the polarisation rotation vanishes, reducing the QBER to 0%. Above 42°C, Δn becomes negative and its magnitude grows, causing the polarisation to scramble again and the QBER to rise to 14% at 60°C. This behaviour, while exaggerated by the model’s temperature coefficient, illustrates the interaction between environmental temperature and polarisation-dependent QBER.

Panel D: QBER vs PMD coefficient. The QBER at 75 km increases linearly with the PMD coefficient from 0.6% at 0 ps/ $\sqrt{\text{km}}$ to 4.4% at 0.3 ps/ $\sqrt{\text{km}}$, consistent with the expected PMD penalty for a 30 ps pulse.

Panel E: QBER vs bend radius. Tight bends ($R < 10$ mm) produce strong bend-induced birefringence $\Delta n_{\text{bend}} \propto (r_{\text{clad}}/R)^2$, reducing the characteristic length L_{char} to below 1 km and scrambling the polarisation fully (QBER $\approx 70\%$). As the bend radius increases beyond 20 mm, the bend contribution becomes negligible compared to Δn_0 , and the QBER returns to the distance-dependent baseline (19% at 75 km with a 50 mm bend).

Combined impairment scenarios (Table 10). At 100 km with 30 ps pulses, the birefringence-dominated QBER is 43% across all scenarios that include fibre transmission, with the tight-bend scenario ($R = 1$ cm) producing 70%. The CD-only and PMD-only scenarios match the all-impairments scenario because the narrow-band 30 ps pulse is insensitive to CD and PMD at this distance, while the birefringence dominates the error rate. For scenarios using shorter pulses, the CD and PMD contributions would be significantly larger, as shown in Panel B.

The five panels together confirm that the independently validated components compose correctly and produce physically sensible behaviour across the full parameter space. The smooth QBER curves, the diffusive transition with distance, the V-shaped temperature response, and the bend-radius dependence are all consistent with the underlying physical models validated in Section 4.

5 Discussion

5.1 Summary of Validation Results

5.2 Parameter Sensitivity

The default fibre parameters are within published specifications for Corning SMF-28 Ultra. The attenuation coefficient $\alpha = 0.182$ dB/km is within the SMF-28e range (0.19–0.22 dB/km) and slightly above the Ultra max spec (0.18 dB/km) — a conservative choice. The residual birefringence $\Delta n_0 = 8.7 \times 10^{-6}$ corresponds to a beat length $L_B \approx 0.18$ m, between standard SM ($L_B > 10$ m) and PM ($L_B < 1$ cm) fibre, which is appropriate for the moderate birefringence in spooled fibre.

5.3 Reproducibility

All simulations are reproducible:

- Seeded RNG (- -seed N, default 42) pinned at session start
- All validation scripts produce seed-tagged CSV + PNG outputs

Table 10: Summary of all component validations.

Component	Reference	Metric	Result
CWLASer (power)	Henry, Coldren	$ P_{\text{meas}} - P_{\text{theory}} /P_{\text{theory}}$	$< 5 \times 10^{-4}\%$
CWLASer (phase)	Henry Eq. 18	$\sum \Delta\phi^2$ vs $D_\phi t$	Matches
CWLASer (RIN)	Coldren Eq. 5.3.38	PSD shape	Peak at 5 GHz, roll-off
MZM (transfer)	Agrawal §4.2	$\cos^2(\pi V/2V_\pi)$	Verified
MZM (chirp)	Koyama and Iga	$\phi_{\text{pp}} = 0, \phi_{\text{sd}} = \pi V/2V_\pi$	Verified
CD	Agrawal Eq. 2.4.11	$\sigma(z)/\sigma_0$	$< 10^{-13}\%$
PMD	Razavi Fig. 2.11	KS test vs Maxwellian	$p = 0.42$
Attenuation	Keiser Eq. 3.6	$P(L)/P(0)$	$< 10^{-13}\%$
Birefringence	Menyuk & Wai, Ulrich Eq. 1	Rotation vs $L, \Delta n(T, R)$	PASS
APD (R)	Kasap Eq. 4.19	$R = \eta e\lambda/hc$	Verified
APD (noise)	Kasap Eq. 4.45	F on shot terms only	Verified

- `run_all.py` executes all validations in subprocesses (isolates module-level side effects)
- 48 unit tests (pytest) with automated seed pinning in `conftest.py`
- Version-controlled repository with tagged commits

5.4 Limitations

The current model has several limitations that represent opportunities for future work:

1. Phenomenological birefringence: The random-axis model uses a phenomenological characteristic length L_{char} calibrated to produce a diffusive transition over 10–150 km. A first-principles multi-section model with realistic fibre beat lengths ($L_B \sim 10\text{--}100$ m for standard SMF-28, compared to $L_B \approx 0.18$ m for our Δn_0) would resolve the random walk over the true correlation length. Our phenomenological approach captures the essential physics — diffusive polarisation scrambling on the Poincaré sphere — while keeping the computation tractable for system-level QKD simulation.
2. No polarisation-dependent loss (PDL): Real fibres and in-line components have PDL that interacts with PMD.
3. No Kerr nonlinearity: At powers > 0 dBm and distances > 50 km, self-phase modulation and four-wave mixing become relevant. The current model is valid for typical QKD powers (−20 to 0 dBm).
4. Single-segment PMD: The PMD vector is drawn once per `cable()` call. A multi-segment model would allow frequency correlation and second-order PMD.
5. CWLASer phase noise is pure Wiener: No $1/f$ phase noise component. Henry [16] describes only white frequency noise; adding $1/f$ noise would increase the linewidth at short timescales.
6. Fixed dispersion parameters: The model currently uses a single D value; a full Sellmeier model would allow wavelength-dependent dispersion over broad bandwidths.
7. No temperature-dependent attenuation: The attenuation coefficient varies slightly with temperature in deployed fibres.

6 Conclusion

We have presented an open-source, physically validated optical channel framework for QKD simulation. Five active components – CW laser, Mach-Zehnder modulator, fibre channel (birefringence, CD, PMD, attenuation), and avalanche photodiode – are independently validated against published literature. The primary contribution is the validation methodology: every equation cites a specific numbered reference, and every component is verified independently, providing a level of traceability absent from existing quantum network and optical system simulators.

The system-level BB84 demonstration confirms that the independently validated components compose correctly, producing physically sensible QBER under combined CD+PMD. This demonstration serves as evidence of correct composition rather than as a standalone QKD contribution; the framework’s value lies in enabling defensible studies of environmental sensitivity and physical-layer security that existing QKD simulators cannot support.

The framework is available under an open-source licence at <https://github.com/azaan-mahmood/o>

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